Introductory Econometrics

Multivariate Regression Model

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Multiple Regression Model

- **Contents**
  - Consistency of OLS
  - Asymptotic Normality and Large-Sample Inference
  - Units of Measurement
  - More on Functional Form
  - More on Goodness-of-Fit and Selection of Regressors

Multiple Regression Model

- **Classical linear model assumptions in MLR:**
  - MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
  - MLR.2: random sampling from the population
  - MLR.3: no perfect collinearity in the sample
  - MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
  - MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)
  - MLR.6: the distribution of u given $(x_1, ..., x_k)$ is $Normal(0, \sigma^2)$

Multiple Regression Model

- **So far we focused on properties of OLS that hold for any sample**
- **Properties of OLS that hold for any sample/sample size**
  - Expected values/unbiasedness under MLR.1 – MLR.4
  - Variance formulas under MLR.1 – MLR.5
  - Gauss-Markov Theorem under MLR.1 – MLR.5
  - Exact sampling distributions/tests under MLR.1 – MLR.6
- **Properties of OLS that hold in large samples**
  - Consistency under MLR.1 – MLR.4
  - Asymptotic normality/tests under MLR.1 – MLR.5



Without assuming normality of the error term!

Consistency of OLS

- **Consistency**

  An estimator θ_n (based on a sample of size n) is consistent for a population parameter θ if

  $$P\left(|\theta_n - \theta| < \epsilon\right) \rightarrow 1 \quad \text{for every } \epsilon > 0 \text{ and } n \rightarrow \infty .$$

  Alternative notation: $plim\ \theta_n = \theta$ ← The estimate converges in probability to the true population value

- **Interpretation:**
  - Consistency means that the probability that the estimate is arbitrarily close to the true population value can be made arbitrarily high by increasing the sample size

- **Example: Sample mean**

  $$plim(\bar{x}_n) = \mu$$ ← Sample mean is a consistent estimator of the population mean

Law of Large Number

Law of Large Numbers (LLN): If $Y_1, \ldots, Y_n$ are i.i.d. and $\sigma_Y^2 < \infty$, then $\bar{Y}$ is a consistent estimator of μ_Y, that is, for any $\epsilon > 0$,

$$P(|\bar{Y} - \mu_Y| < \epsilon) \to 1 \text{ as } n \to \infty .$$

It is written as $\bar{Y} \xrightarrow{p} \mu_Y$, for which we can read "$\bar{Y}$ converges in probability to μ_Y".

- It can be proved by the Chebyshev inequality, since

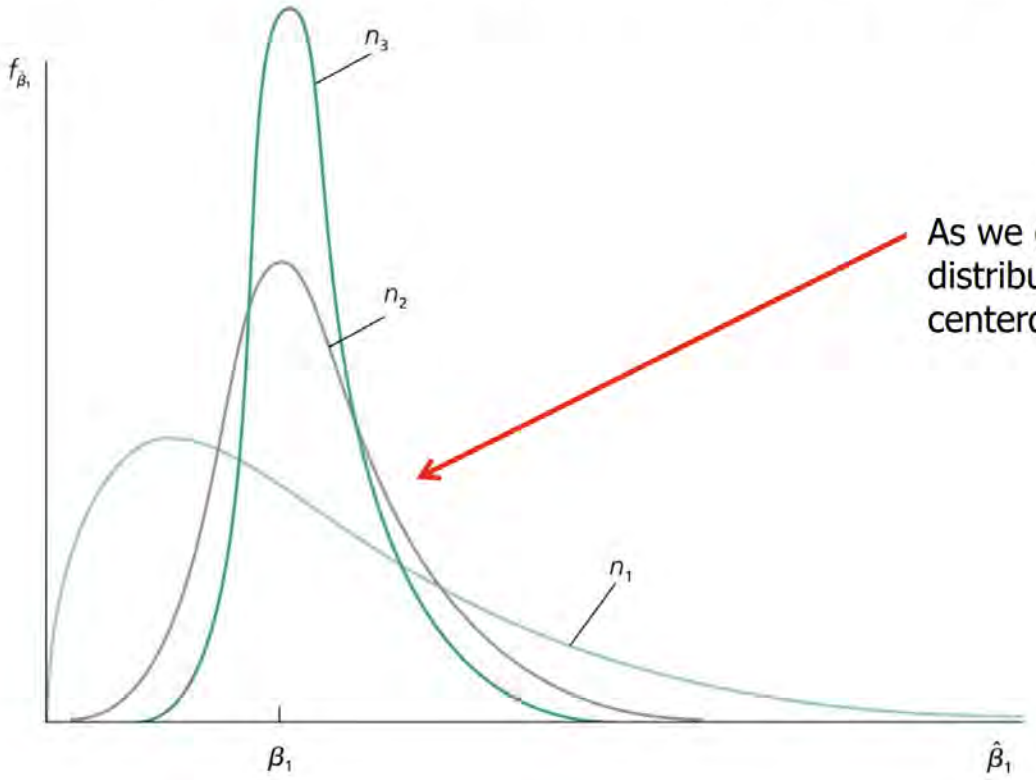
$$\begin{aligned} var(\bar{Y}) &= var(\frac{1}{n}\sum_{i=1}^{n} Y_i) = \frac{1}{n^2} var(\sum_{i=1}^{n} Y_i) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} cov(Y_i, Y_j) = \frac{1}{n^2}\sum_{i=1}^{n} var(Y_i) \\ &= \frac{\sigma_Y^2}{n} \to 0 \text{ as } n \to \infty \text{ (i.i.d.)} \end{aligned}$$

Consistency of OLS

- **Theorem 5.1 (Consistency of OLS)**

$$MLR.1\text{–}MLR.4 \quad \Rightarrow \quad plim\ \hat{\beta}_j = \beta_j, \quad j = 0, 1, \ldots, k$$

- Sampling distribution of $\hat{\beta}_1$ for $n_1 < n_2 < n_3$



As we get more data, the samping distribution of $\hat{\beta}_1$ is more tighly centerd around β_1

Consistency of OLS

- **Special case of simple regression model**

$$\hat{\beta}_1 = \beta_1 + \frac{n^{-1}\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)u_i}{n^{-1}\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

$$plim\ \hat{\beta}_1 = \beta_1 + \frac{Cov(x_1, u)}{Var(x_1)}$$

One can see that the slope estimate is consistent if the explanatory variable is exogenous, i.e. uncorrelated with the error term.

- **Assumption MLR.4'**

$$E(u) = 0$$

$$Cov(x_j, u) = 0$$

All explanatory variables must be uncorrelated with the error term. This assumption is weaker than the zero conditional mean assumption MLR.4.

Consistency of OLS

- **For consistency of OLS, only the weaker MLR.4` is needed**
- **Asymptotic analog of omitted variable bias**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + v$$ ← True model

$$y = \beta_0 + \beta_1 x_1 + [\beta_2 x_2 + v] = \beta_0 + \beta_1 x_1 + u$$ ← Misspecified model

$$\Rightarrow \quad plim\ \tilde{\beta}_1 = \beta_1 + Cov(x_1, u)/Var(x_1)$$

$$= \beta_1 + \beta_2 Cov(x_1, x_2)/Var(x_1) = \beta_1 + \beta_2 \delta_1$$ ← Bias

There is no omitted variable bias if the omitted variable is irrelevant or uncorrelated with the included variable

Asymptotic Normality and Large-Sample Inference

- **Asymptotic normality and large sample inference**
  - In practice, the normality assumption MLR.6 is often questionable
  - If MLR.6 does not hold, the results of t- or F-tests may be wrong
  - Fortunately, F- and t-tests still work if the sample size is large enough
  - Also, OLS estimates are normal in large samples even without MLR.6
- **Theorem 5.2 (Asymptotic normality of OLS)**

Under assumptions MLR.1 – MLR.5:

$$\frac{(\hat{\beta}_j - \beta_j)}{se(\hat{\beta}_j)} \overset{a}{\sim} N(0,1)$$

In large samples, the standardized estimates are normally distributed

Asymptotic Normality and Large-Sample Inference

- **Practical consequences**
  - In large samples, the t-distribution is close to the N(0,1) distribution
  - As a consequence, t-tests are valid in large samples without MLR.6
  - The same is true for confidence intervals and F-tests
  - Important: MLR.1 – MLR.5 are still necessary, esp. homoscedasticity
- **Asymptotic analysis of the OLS sampling errors**

$$\widehat{Var}(\hat{\beta}_j) = \frac{\hat{\sigma}^2}{SST_j(1-R_j^2)}$$

Converges to σ^2

Converges to $n \cdot Var(x_j)$

Converges to a fixed number

Asymptotic Normality and Large-Sample Inference

- **Asymptotic analysis of the OLS sampling errors (cont.)**

  $\widehat{Var}(\hat{\beta}_j)$ shrinks with the rate $1/n$

  $se(\hat{\beta}_j)$ shrinks with the rate $\sqrt{1/n}$

- **<u>This is why large samples are better</u>**
- **Example: Standard errors in a birth weight equation** (BWGHT.dta)

$$log(bwght) = \beta_0 + \beta_1 cigs + \beta_2 log(faminc)$$

$n = 1,388 \Rightarrow se(\hat{\beta}_{cigs}) = .00086$

$n = 694 \Rightarrow se(\hat{\beta}_{cigs}) = .0013$

$$\frac{.00086}{.0013} \approx \sqrt{\frac{694}{1,388}}$$

Use only the first half of observations

Units of Measurement

- **Effects of Data Scaling on OLS Statistics**
- **Example: Birth weight equation**

  Data: BWGHT.dta

Birth weight, in ounces

Number of cigarettes the mother smoked per day

Annual family income, in thousands of dollar

$$bwght = \beta_0 + \beta_1 cigs + \beta_2 faminc + u$$

- What if we measure birth weight in pounds, rather than in ounces?

$$bwghtlbs = bwght/16$$

- What if we measure smoking by number of packs of cigarettes per day?

$$packs = cigs/20$$

Units of Measurement

- **Effects of Data Scaling**

Dependent Variable	(1) *bwght*	(2) *bwghtlbs*	(3) *bwght*
Independent Variables			
cigs	−.4634 (.0916)	−.0289 (.0057)	—
packs	—	—	−9.268 (1.832)
faminc	.0927 (.0292)	.0058 (.0018)	.0927 (.0292)
intercept	116.974 (1.049)	7.3109 (.0656)	116.974 (1.049)
Observations	1,388	1,388	1,388
R-Squared	.0298	.0298	.0298
SSR	557,485.51	2,177.6778	557,485.51
SER	20.063	1.2539	20.063

Units of Measurement

- **Beta Coefficients**

  One can standardize all variables in a regression model:

$$\frac{(y_i - \bar{y})}{\hat{\sigma}_y} = (\frac{\hat{\sigma}_1}{\hat{\sigma}_y})\hat{\beta}_1(\frac{x_{i1} - \bar{x}_1}{\hat{\sigma}_1}) + \ldots + (\frac{\hat{\sigma}_k}{\hat{\sigma}_y})\hat{\beta}_k(\frac{x_{ik} - \bar{x}_k}{\hat{\sigma}_k}) + \frac{\hat{u}_i}{\hat{\sigma}_y}$$

  Alternative notation: Beta coefficents

$$z_y = \hat{b}_1 z_1 + \hat{b}_2 z_2 + \ldots + \hat{b}_k z_k + error$$

$$\hat{b}_j = (\frac{\hat{\sigma}_j}{\hat{\sigma}_y})\hat{\beta}_j \quad \text{for} \quad j = 1, \ldots, k$$

- **Interpretation**: If x_j increases by one standard deviation, then y changes by $\hat{b}_j$ standard deviations.

Units of Measurement

- Beta coefficient is useful when some of the independent variables or dependent variable have units that are not easily understood
- Beta coefficient is helpful in determining the economic significance
- Beta coefficients can be used to compare the effects among independent variables
- **Example: Class attendance and final exam score**

Data: ATTEND.DTA

$$\widehat{final} = 12.37 - .079\ missed + 1.915\ priGPA + .401\ ACT$$

How large is the effect? Is it economically significant?
(It dependends on the variation of final exam score)

$1.915 > 0.401$

Does it mean priGPA has a larger effect on final score than ACT does?
(It dependends on the variation of priGPA and ACT)

Units of Measurement

- **Example: Effects of pollution on housing prices**

  Data: HPRICE2.dta



House price, in dollars

Nitrogen oxide in the air, parts per 100m

$$price = \beta_0 + \beta_1 nox + \beta_2 crime + \beta_3 rooms + \beta_4 dist + \beta_5 stratio + u$$

- One way to understand the size of the air quality is to compute beta coefficient:

$$\widehat{zprice} = -.34 znox - .14 zcrime + .51 zrooms - .24 zdist - .27 zstratio$$

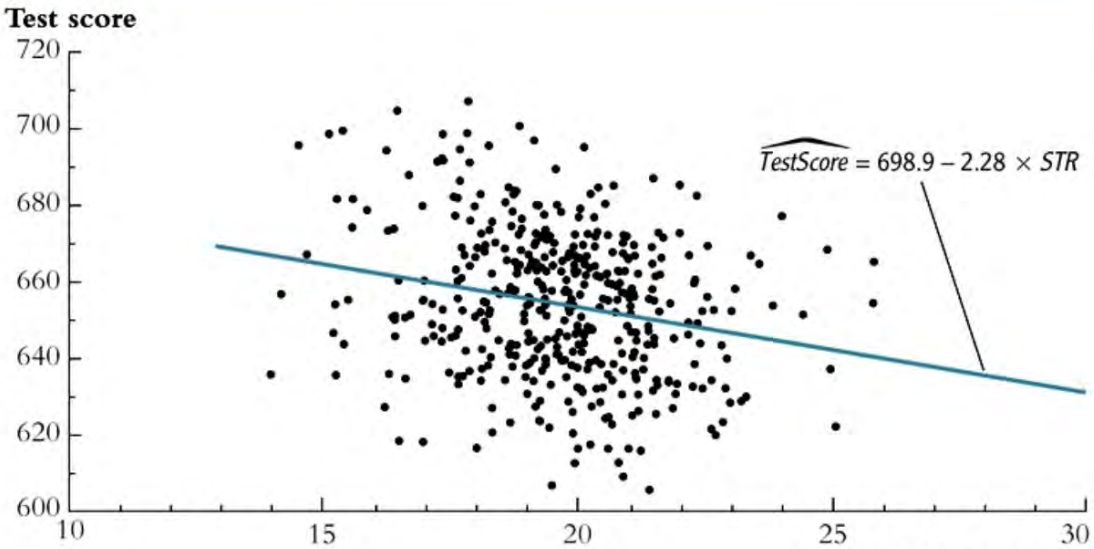
One standard deviation increase in nox decreases the price by 0.34 standard deviation

More on Functional Form

Empirical example: Class size and educational output

FIGURE 4.3 The Estimated Regression Line for the California Data

The estimated regression line shows a negative relationship between test scores and the student-teacher ratio. If class sizes fall by 1 student, the estimated regression predicts that test scores will increase by 2.28 points.

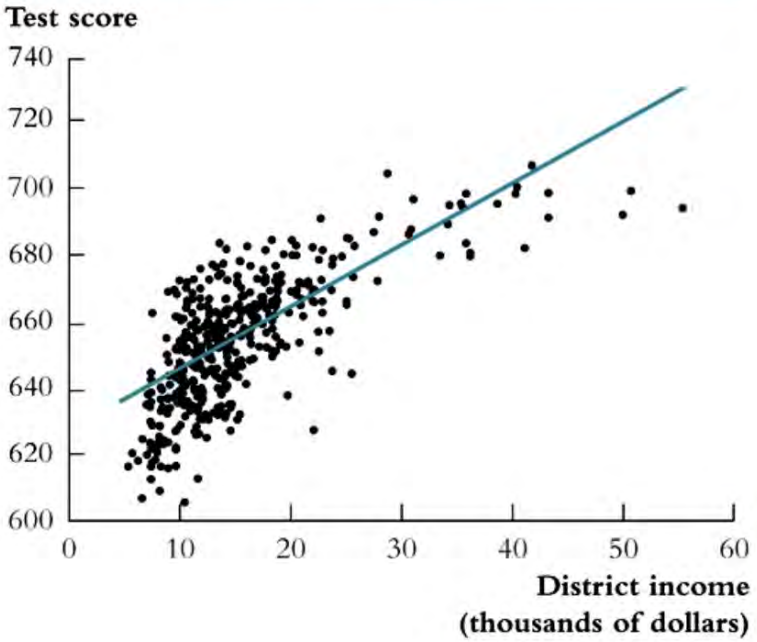


- The TestScore – STR relation looks approximately linear . . .

More on Functional Form

FIGURE 6.2 **Scatterplot of Test Score vs. District Income with a Linear OLS Regression Function**

There is a positive correlation between test scores and district income (correlation = 0.71), but the linear OLS regression line does not adequately describe the relationship between these variables.



- But the TestScore – Average District Income relation looks nonlinear

More on Functional Form

If a relation between Y and X is nonlinear:

- The effect on Y of a change in X depends on the value of X – that is, the marginal effect of X is not constant
- A linear regression is misspecified – the functional form is wrong
- The estimator of the effect of X is biased – it needn't even be right on average
- The solution to this is to estimate a regression function that is nonlinear in X

More on Functional Form

- Polynomial Functions
- Logarithmic Transformations
- Interaction between Independent Variables

More on Functional Form

- $\ln(X)$ = the natural logarithm of X
- Logarithmic transformation is useful in modeling relations in "percentage" terms

Why percentage?

$$\ln(x + \Delta x) - \ln(x) = \ln\left(1 + \frac{\Delta x}{x}\right) \approx \frac{\Delta x}{x},$$

because $\ln(1 + z) \approx z$ when z is small by the Taylor's expansion.

- $\ln(1.01) = 0.00995 \approx 0.01$
- $\ln(1.10) = 0.0953 \approx 0.10$ (sort of)

More on Functional Form

Three cases:

Linear-log: $Y = \beta_0 + \beta_1 \ln(X) + u$

Log-linear: $\ln(Y) = \beta_0 + \beta_1 X + u$

Log-log: $\ln(Y) = \beta_0 + \beta_1 \ln(X) + u$

- The interpretation of the slope coefficient differs in each case
- The interpretation is found by applying the general "before and after" rule: "figure out the change in Y for a given change in X."

More on Functional Form

Linear-log:

$$Y = \beta_0 + \beta_1 \ln(X) + u$$

For small ΔX,

$$\beta_1 \approx \frac{\Delta Y}{\Delta X / X} .$$

- A 1% increase in X is associated with a $0.01\beta_1$ unit increase in Y
- 1% increase in $X \Rightarrow$ 0.01 unit increase in $\ln(X) \Rightarrow 0.01\beta_1$ unit increase in Y

More on Functional Form

Empirical example: TestScore - Income relation

- First defining the new regressor, ln(Income)
- The model is now linear in ln(Income), so the linear-log model can be estimated by OLS:

$$\widehat{\text{TestScore}}_i = \underset{(3.8)}{557.8} + \underset{(1.40)}{36.42} \times \ln(\text{Income}_i) \, .$$

  So a 1% increase in Income is associated with an increase in TestScore of 0.36 points on average

- Standard errors, confidence intervals, R^2-all the usual tools of regression apply here
- How does this compare to the cubic model?

More on Functional Form

Log-linear:

$$\ln(Y) = \beta_0 + \beta_1 X + u$$

For small ΔX,

$$\beta_1 \approx \frac{\Delta Y / Y}{\Delta X} .$$

- 1 unit increase in X is associated with a $100\beta_1\%$ increase in Y
- 1 unit increase in $X \Rightarrow \beta_1$ unit increase in $\ln(Y) \Rightarrow 100\beta_1\%$ increase in Y

More on Functional Form

Log-log:

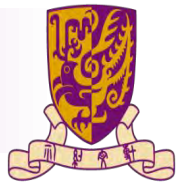
$$\ln(Y) = \beta_0 + \beta_1 \ln(X) + u$$

For small ΔX,

$$\beta_1 \approx \frac{\Delta Y/Y}{\Delta X/X}.$$

- A 1% increase in X is associated with a β_1% increase in Y
- 1% increase in $X \Rightarrow$ 0.01 unit increase in $\ln(X) \Rightarrow 0.01\beta_1$ unit increase in $\ln(Y) \Rightarrow \beta_1$% increase in Y
- In the log-log specification, β_1 has the interpretation of an elasticity

More on Functional Form



Summary

- Three cases, differing in whether Y and/or X is transformed by taking logarithms
- The regression is linear in the new variable(s) $\ln(Y)$ and/or $\ln(X)$, and the coefficients can be estimated by OLS
- Hypothesis tests and confidence intervals are now implemented and interpreted "as usual"
- The interpretation of β_1 differs from case to case
- Choice of specification should be guided by judgment (e.g., which interpretation makes the most sense in your application?), tests, and plotting predicted values

More on Functional Form

- **Reasons for using the natural log**
  - Convenient percentage/elasticity interpretation
  - Slope coefficients of logged variables are invariant to rescalings
  - Taking logs often eliminates/mitigates problems with outliers
  - Taking logs often helps to secure normality and homoscedasticity
- **Limitations of using the natural log**
  - Logs must not be used if variables take on zero or negative values
  - It is hard to reverse the log-operation when constructing predictions
- **General rules of using the natural log**
  - Logs are offen used for dollar amounts that are always positive
  - Variables measured in years should not be logged
  - Variables measured in percentage points should also not be logged

More on Functional Form

- **Using quadratic functional forms**

  Models with quadratics can capture increasing or decreasing effects

- **Example: Wage equation**

  Data: WAGE1.dta

  Concave experience profile

$$\widehat{wage} = \underset{(.35)}{3.73} + \underset{(.041)}{.298}\, exper - \underset{(.0009)}{.0061}\, exper^2$$

$$n = 526, R^2 = .093$$

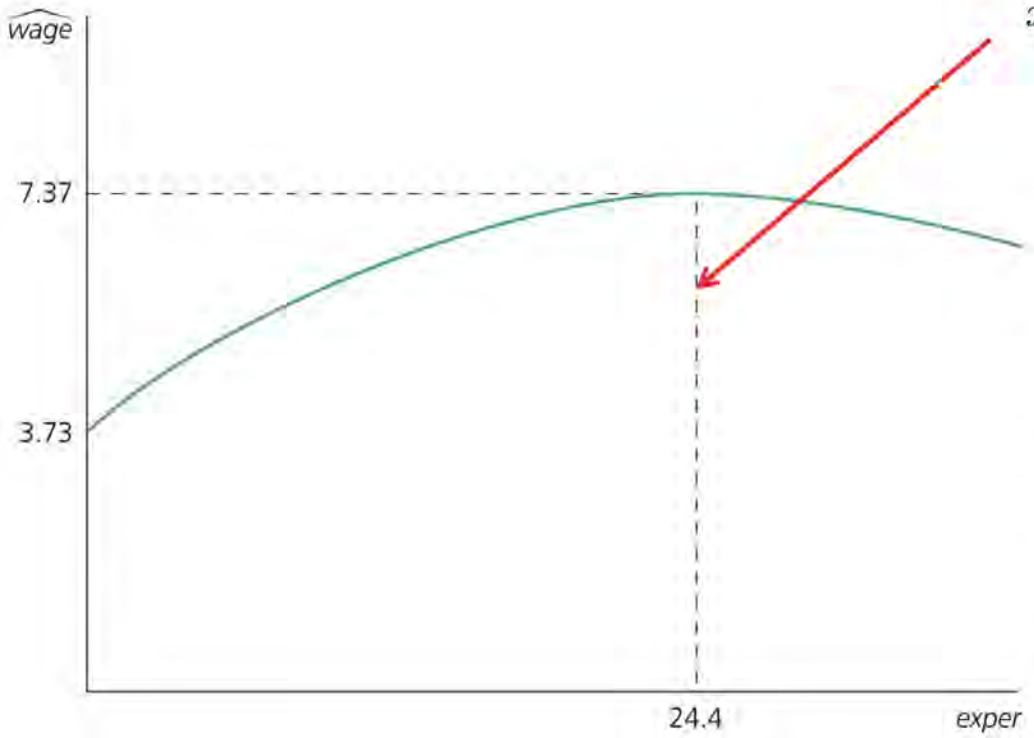
- **Marginal effect of experience**

  The first year of experience increases the wage by some .30$, the second year by .298-2(.0061)(1) = .29$ etc.

$$\frac{\partial wage}{\partial exper} = .298 - 2(.0061)exper$$

More on Functional Form

- **Wage maximum with respect to work experience**



$$x^* = \left|\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right| = \left|\frac{.298}{2(.0061)}\right| \approx 24.4$$

Does this mean the return to experience becomes negative after 24.4 years?

Not necessarily. It depends on how many observations in the sample lie right of the turnaround point.

In the given example, these are about 28% of the observations. There may be a specification problem (e.g. omitted variables).

More on Functional Form

- **Example: Effects of pollution on housing prices**

Data: HPRICE2.dta

Nitrogen oxide in air, distance from employment centers, student/teacher ratio

$$\widehat{\log(price)} = \underset{(.57)}{13.39} - \underset{(.115)}{.902}\ \log(nox) - \underset{(.043)}{.087}\ \log(dist)$$

$$- \underset{(.165)}{.545}\ rooms + \underset{(.013)}{.062}\ rooms^2 - \underset{(.006)}{.048}\ stratio$$

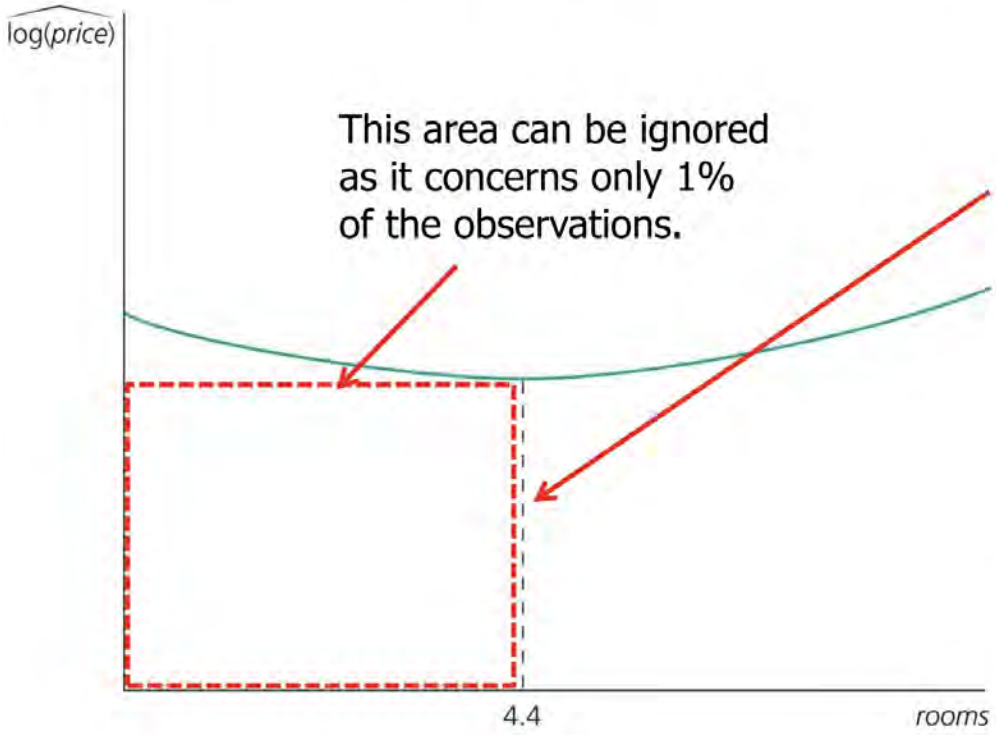
$$n = 506,\ R^2 = .603$$

Does this mean that, at a low number of rooms, more rooms are associated with lower prices?

$$\Rightarrow \quad \frac{\partial \log(price)}{\partial rooms} = -.545 + .124\, rooms$$

More on Functional Form

- **Calculation of the turnaround point**



Turnaround point:

$$x^* = \left| \frac{-.545}{2(.062)} \right| \approx 4.4$$

Increase rooms from 5 to 6:

$$-.545 + .124(5) = +7.5\% \ price$$

Increase rooms from 6 to 7:

$$-.545 + .124(6) = +19.9\% \ price$$

More on Functional Form

- **Other possibilities**

$$\log(price) = \beta_0 + \beta_1 \log(nox) + \beta_2 [\log(nox)]^2$$

$$+ \beta_3 crime + \beta_4 rooms + \beta_5 rooms^2 + \beta_6 stratio + u$$

$$\Rightarrow \quad \frac{\partial \log(price)}{\partial \log(nox)} = \frac{\%\partial price}{\%\partial nox} = \beta_1 + 2\beta_2 [\log(nox)]$$

The elasticity of $price$ with respect to nox now depends on $\log(nox)$

- **Higher polynomials**

$$cost = \beta_0 + \beta_1 quantity + \beta_2 quantity^2 + \beta_3 quantity^3 + u$$

More on Functional Form

- **Models with interaction terms**

$$\log(price) = \beta_0 + \beta_1 sqrft + \beta_2 bdrms + \beta_3 sqrft \cdot bdrms + \beta_4 bthrms + u$$

Interaction term

$$\Rightarrow \quad \frac{\partial \log(price)}{\partial bdrms} = \beta_2 + \beta_3 sqrft$$

The effect of the number of bedrooms depends on the level of square footage

- **How to test whether there is an interaction effect?**
- **Interaction effects complicate interpretation of parameters**

$\beta_2 =$ Effect of number of bedrooms, but for a square footage of zero

Impractical !!

More on Functional Form

- **Reparametrization of interaction effects**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$$

sample means

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u$$

New interaction term

$$\Rightarrow \quad \frac{\partial \log(y)}{\partial x_2} = \delta_2 + \beta_3 (x_1 - \mu_1)$$

Effect of x_2 if x_1 take on its mean value

- **Advantages of reparametrization**
  - Easy interpretation of all parameters
  - Standard errors for partial effects at the mean values available
  - If necessary, interaction may be centered at other interesting values

More on Goodness-of-Fit and Selection of Regressors

- **More on goodness-of-fit and selection of regressors**
- **General remarks on R-squared**
  - A high R-squared does not imply that there is a causal interpretation
  - A low R-squared does not preclude precise estimation of partial effects
  - R-squared never goes down when one or more variables are added to the regression
- **Adjusted R-squared**
  - What is the ordinary R-squared supposed to measure?

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{(SSR/n)}{(SST/n)}$$ is an estimate for $$1 - \frac{\sigma_u^2}{\sigma_y^2}$$

Population R-squared

More on Goodness-of-Fit and Selection of Regressors

- **Adjusted R-squared (cont.)**

Correct degrees of freedom of nominator and denominator

  - A alternative estimate taking into account degrees of freedom would be

$$\bar{R}^2 = 1 - \frac{(SSR/(n-k-1))}{(SST/(n-1))} = adjusted\ R^2$$

  - The adjusted R-squared imposes a penalty for adding new regressors
  - The adjusted R-squared increases if, and only if, the t-statistic of a newly added regressor is greater than one in absolute value

- **Relationship between R-squared and adjusted R-squared**

$$\bar{R}^2 = 1 - (1 - R^2)\frac{(n-1)}{(n-k-1)}$$

The adjusted R-squared may even get negative

More on Goodness-of-Fit and Selection of Regressors

- **Using adjusted R-squared to choose between nonnested models**

  Models are **nonnested** if neither model is a special case of the other

- **Example: R&D intensity and firm sales** (RDCHEM.dta)

$$rdintens = \beta_0 + \beta_1 \log(sales) + u \quad \leftarrow \quad R^2 = .061, \bar{R}^2 = .030$$

$$rdintens = \beta_0 + \beta_1 sales + \beta_2 sales^2 + u \quad \leftarrow \quad R^2 = .148, \bar{R}^2 = .090$$

  - Both models could capture a diminishing effect
  - A comparison between the R-squared of both models would be unfair to the first model because the first model contains fewer parameters
  - In the given example, even after adjusting for the difference in degrees of freedom, the quadratic model is preferred

More on Goodness-of-Fit and Selection of Regressors

- **Example: Factors Affecting MBL Salaries (See Week 6 slides)**

  There is a multicollinearity between $rbisyr$ and $hrunsyr$. Which variable should be dropped?

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + u$$

$$\bar{R}^2 = .6211$$

$rbisyr$ is droped from the model

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 rbisyr + u$$

$$\bar{R}^2 = .6226$$

$hrunsyr$ is droped from the model

There is a slight preference for the model with $rbisyr$.

More on Goodness-of-Fit and Selection of Regressors

- **Comparing models with different dependent variables**
  - R-squared or adjusted R-squared must not be used to compare models which differ in their definition of the dependent variable
- **Example: CEO compensation and firm performance**

  Data: CEOSAL1

$$\widehat{salary} = \underset{(223.63)}{223.90} + \underset{(.0163)}{.0089}\, sales + \underset{(11.08)}{19.63}\, roe$$

$$n = 209, R^2 = .029, \bar{R}^2 = .020, TSS = 391,732,982$$

There is much less variation in log(salary) that needs to be explained than in salary

$$\widehat{lsalary} = \underset{(0.29)}{4.36} + \underset{(.033)}{.275}\, lsales + \underset{(.0040)}{.0179}\, roe$$

$$n = 209, R^2 = .282, \bar{R}^2 = .275, TSS = 66.72$$

More on Goodness-of-Fit and Selection of Regressors

- **Controlling for too many factors in regression analysis**
- **In some cases, certain variables should not be held fixed**
  - In a regression of traffic fatalities on state beer taxes (and other factors) one should not directly control for beer consumption
  - In a regression of family health expenditures on pesticide usage among farmers one should not control for doctor visits
- **Different regressions may serve different purposes**
  - In a regression of house prices on house characteristics, one would only include price assessments if the purpose of the regression is to study their validity; otherwise one would not include them

More on Goodness-of-Fit and Selection of Regressors

- **Adding regressors to reduce the error variance**
  - Adding regressors may excarcerbate multicollinearity problems
  - On the other hand, adding regressors reduces the error variance
  - Variables that are uncorrelated with other regressors should be added because they reduce error variance without increasing multicollinearity
  - However, such uncorrelated variables may be hard to find
- **Example: Individual beer consumption and beer prices**
  - Including individual characteristics in a regression of beer consumption on beer prices leads to more precise estimates of the price elasticity